MIT 18.06SC - 3.0

Problem 23.1

$$\frac{d}{dt}\|u(t)\|^2 = \frac{d}{dt}(u_1^2 + u_2^2 + u_3^2)$$

$$= 2u_1u_1' + 2u_2u_2' + 2u_3u_3'$$

$$= 2u_1(cu_2 - bu_3) + 2u_2(au_3 - cu_1) + 2u_3(bu_1 - au_2)$$

$$= 2u_1u_2c - 2u_1u_3b + 2u_2u_3a - 2u_2u_1c + 2u_3u_1b - 2u_3u_2a$$

$$= 0$$

Thus the length of u is constant and not changing i.e. $\|u(t)\|^2 = = u(0)$ for every value of t ~~and since $\|u(t)\|^2$ does not change~~

Problem 23.2

$(1-\lambda)(3-\lambda) = 0$ ~~when~~ and so $\lambda_1 = 1$ and $\lambda_2 = 3$

$$\begin{bmatrix} 0 & 1 \\ 0 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} + \begin{bmatrix} 1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

So $x_2 = 0$ and ~~eigenvecto~~ $A\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ = eigenvector of $\lambda_1 = 1$ is $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$$\begin{bmatrix} -2 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} x_1 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving for x_1 gives $1/2$ so an eigenvector of $\lambda_2 = 3$ is $2\cdot\begin{bmatrix} 1/2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$

$$S\Lambda S^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}\frac{1}{2}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$Se^{\Lambda t}S^{-1} = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}\begin{bmatrix} e^t & 0 \\ 0 & e^{3t} \end{bmatrix}\frac{1}{2}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & e^{3t} \\ 0 & 2e^{3t} \end{bmatrix}\frac{1}{2}\begin{bmatrix} 2 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} e^t & -\frac{1}{2}e^t + \frac{1}{2}e^{3t} \\ 0 & e^{3t} \end{bmatrix} = e^{At}$$

$$e^{A(0)} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I_2$$

$$\frac{d}{dt}e^{At} = \begin{bmatrix} e^t & -\frac{1}{2}e^t + \frac{3}{2}e^{3t} \\ 0 & 3e^{3t} \end{bmatrix}$$

$$\frac{d}{dt}e^{At}\Big|_{t=0} = \begin{bmatrix} e^0 & -\frac{1}{2}e^0 + \frac{3}{2}e^{3(0)} \\ 0 & 3e^{3(0)} \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & 3 \end{bmatrix} = A$$